

Hackathon Solution Verification: Content & Design Engine

1. GitHub Repository

<https://github.com/aviraltrivedi/content-design-engine> (<https://github.com/aviraltrivedi/content-design-engine>)

2. System Architecture

```
graph TD
  A[Browser Dashboard] -->|Upload Files| B[Express Server]
  B -->|Save to| C[./input Directory]
  C -->|Chokidar Watcher| D[Design Engine]
  D -->|Score Asset| E{Quality > 10?}
  E -->|No| F[./low-confidence Directory]
  E -->|Yes| G[Debounce & Batch Queue]
  G -->|Trigger Pipeline| H[Transformer & Documenter]
  H -->|Move original assets| I[./processed Directory]
  I -->|Generate Images/Docs| J[./output Directory]
  J -->|AJAX Polling| A
```

3. Input-to-Output Flow Verification

Upload & Real-Time Scoring

Assets are uploaded directly via the interactive dashboard and appear in the `Live Scoring Pipeline` section as they are autonomously evaluated by the system.

(Logs showing successful detection and scoring)

```
📄 New asset detected: 1777309397009-SUP_8294.JPG
✅ Quality approved. Queuing transformation...
📄 New asset detected: 1777309397410-SUP_8297.JPG
✅ Quality approved. Queuing transformation...
```

Output Generation

After the debounce buffer collects the batch (e.g., 5 images), it triggers the transformation. The original assets are moved to `/processed` to prevent duplicates, and the final collages and Case Study files are generated in `/output` and displayed live on the dashboard.

4. Asset Selection Logic

The engine employs a real-time, threshold-based autonomous selection logic:

- **Detection:** Assets dropped into `./input` are immediately detected via file system events.
- **Scoring System:** Each asset is evaluated by calculating a composite `Quality Score` based on simulated metrics (contrast, sharpness, resolution, metadata).
- **Thresholding:** Assets scoring above the confidence threshold are approved for transformation.
- **Archival Routing:** Substandard assets are immediately rerouted to `./low-confidence/` for manual review, ensuring only high-quality assets enter the transformation pipeline.
- **Batch Processing:** Approved assets are queued using a 2.5-second debounce buffer and a concurrency lock. Once the batch is finalized, it is packaged together for dynamic layout construction.